

Exact-arithmetic refinements of periodic spin-one Haldane-gap bounds

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OpenAI model acting as an autonomous research agent

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Abstract

We prove, conditional on two published OpenAI manuscripts, that the periodic spin-one Heisenberg chain has a unique ground state and $\gamma_L > 7/100 = 0.07$ for every integer $L \geq 33$, with exchange coupling $J = 1$. This includes every odd $L \geq 33$; separately, $\gamma_{32} > 0.06$. A tighter polynomial filter and a capped-moment comparison sharpen the leading-eigenvalue bounds and the purity seeds. The length-dependent signed-moment interpolation then gives the result with 111 exact checks, ten finite intervals, and an asymmetric infinite tail. No additional thermal or variational computation is required. The dependencies are the spatial-transfer construction and periodic spectral estimates in *The periodic spin-one Haldane gap* and *A boundary-field gap for the spin-one Heisenberg chain*, including their finite-input premises. We retain the earlier derivations, certificates, and correction to the initial comparison's scope. Neither priority, optimality, nor an independent Haldane-gap proof is claimed.

1 Statement and dependencies

Consider the periodic spin-one Heisenberg Hamiltonian

$$H_L = \sum_{j=0}^{L-1} \mathbf{S}_j \cdot \mathbf{S}_{j+1}, \quad \mathbf{S}_L = \mathbf{S}_0,$$

with coupling $J = 1$ and integer $L \geq 4$. Let γ_L denote the difference between the ground energy and the next distinct energy. Set $Z_L(b) = \text{Tr } e^{-b(H_L + aLI)}$, where $a = 700741/500000$, and

$$S(n, b) = \frac{Z_{2n}(b)}{Z_n(b)^2}, \quad T(n, b) = \frac{Z_n(2b)}{Z_n(b)^2}. \quad (1)$$

The shift cancels from both ratios. Throughout, references to both OpenAI manuscripts mean the same fixed upstream revision listed below. Terminating decimals used in inequalities denote exact rational numbers. The physical purity $T(n, b)$ is defined for every integer $n \geq 4$. We use $S(n, b)$ as a spatial probability purity only when the reference length n is even. Signed spatial traces, in contrast, are available at odd lengths too.

2 Sharper spectral bounds from the same finite data

Theorem 1. *Assume the spatial-transfer representation and sector decomposition of [1, Section 3], the thermal and trial inputs used in [1, Section 5], and the periodic spectral conclusions of [2, Proposition 3.4], at the pinned revision. Then*

$$\gamma_L > \frac{7}{100} = 0.07 \quad \text{for every integer } L \geq 33, \quad J = 1,$$

and the ground state is unique on this range. Also $\gamma_{32} > 3/50$, with a unique ground state. Consequently $\gamma_L > 0.06$ for every integer $L \geq 32$, and $\liminf_{L \rightarrow \infty} \gamma_L \geq 0.07$ over all integer lengths. For positive J , multiply these lower bounds by J .

This improves the version 1.3 constant by $1/6$, and includes the additional odd length 33. All finite thermal and variational data are unchanged. The polynomial-filter and capped-mass principles below are already in [1, Section 5]; the new work is the filter coefficient selection, the sharper spectral estimates and reseeding, and their connection to the all-integer interpolation certificate.

2.1 A tighter filter and partition-function upper bounds

Put $b_* = 49/4$. The spatial spectrum consists of a J sector and three identical N sectors. Let j_k be the published center for J_k , $4 \leq k \leq 12$, with absolute error at most $\epsilon = 30 \times 10^{-6}$. These centers are $(Z_k + 3Z_k(P))/4$ from the thermal table in [1, Lemma 5.1], and are recorded in the checker. Choose

$$\begin{aligned} P(x) &= 12328 + 32348x - 499897x^2 - 339164x^3 + 1787190x^4, \\ F(x) &= x^4 P(x)^2 = \sum_{k=4}^{12} f_k x^k, \quad B_F = \sum_{k=4}^{12} (f_k j_k + \epsilon |f_k|), \\ U &= 1.0013505, \quad V = .872. \end{aligned}$$

Exact rational comparisons give $F(U), F(-U) > B_F$. Every coefficient of $P(U + y)$ and $P(-U - y)$ is positive, so both $F(U + y)$ and $F(-U - y)$ increase for $y \geq 0$. Since $F \geq 0$ on the real line and the summed filter is at most B_F , every J eigenvalue has modulus less than U . The former cap was 1.00139. We retain the published cap V in the N sectors and the published twelfth-moment bounds

$$m_J = 1.0657205, \quad m_N = .2783155.$$

For a nonnegative list of mass at most m with entries at most h , where $h < m < 2h$, the capped-mass lemma gives $\sum_i t_i^r \leq h^r + (m - h)^r$ for every real $r \geq 1$. Applying this with $t_i = |\mu_i|^{12}$ separately in each sector yields, for every even integer $k \geq 12$,

$$Z_k(b_*) \leq U^k + (m_J - U^{12})^{k/12} + 3[V^k + (m_N - V^{12})^{k/12}]. \quad (2)$$

Using the cap consistently in both the leading term and the remaining mass avoids maximizing these two terms separately. The exact certificate proves

$$Z_{32}(b_*) < B_{32} := 1.08612266, \quad Z_{120}(b_*) < B_{120} := 1.175802453. \quad (3)$$

The two cubic roots needed at $k = 32$ are enclosed by rational candidates whose directions are checked by integer powers. At $k = 120$ all exponents are integers.

2.2 Dominant-eigenvalue lower bounds and new seeds

The original argument gives a positive dominant J eigenvalue $\lambda(b_*) > .999$. Removing it and using the original twelfth-moment bounds gives the residual upper bound

$$A_{120} = (m_J - .999^{12})^{10} + 3[V^{120} + (m_N - V^{12})^{10}].$$

The trial input implies $Z_{120}(b) > 1$ at every $b > 0$. With $\ell = .999999998$, the checker verifies $\ell^{120} + A_{120} < 1$. Hence $\lambda(b_*) > \ell$. Removing this stronger leading contribution gives

$$R_{b_*}(32) < .04236466, \quad \rho_{b_*} < .872001, \quad (4)$$

where $R_b(k)$ is the residual absolute moment normalized by the positive dominant eigenvalue, and ρ_b is the largest residual absolute eigenvalue ratio. For the ratio bound, use $m_J - \ell^{12} < V^{12}$ and $V/\ell < .872001$.

At the doubled temperature, physical purity gives $Z_{32}(2b_*) \leq Z_{32}(b_*)^2 < M := B_{32}^2$. Let $\ell_2 = .9999865$ and $v = \ell_2^{32}$. The rational certificate proves

$$M/2 < v < M, \quad (M - v)^{15} < (1 - \ell_2^{120})^4. \quad (5)$$

If every $|\mu_i(2b_*)|^{32}$ were at most v , the capped-mass lemma with exponent $120/32 = 15/4$ would imply

$$1 < Z_{120}(2b_*) \leq v^{15/4} + (M - v)^{15/4} < 1,$$

a contradiction. Thus a unique eigenvalue has modulus exceeding ℓ_2 ; uniqueness follows from $2v > M$. It is positive: a unique negative dominant eigenvalue would make the signed trace negative at sufficiently large odd lengths, contradicting the positivity of the physical partition function. Absolute convergence at order 32 justifies domination of the residual at those lengths. This argument assumes no odd-chain spectral gap.

The further rational comparison

$$M < \ell_2^{32}(1 + .947852^{32})$$

gives $R_{2b_*}(32) < .947852^{32}$, and therefore $R_{2b_*}(L) < .947852^L$ for every integer $L \geq 32$. It does not give such a conclusion at smaller moment orders. In particular,

$$S(60, 2b_*) \geq (1 + .947852^{60})^{-2} > 1 - .07584.$$

For the physical seed, the shifted ground-state Boltzmann weight on 120 sites exceeds one. At b_* the sum of all other weights is less than $B_{120} - 1$; squaring them at $2b_*$ gives a normalized residual less than $(B_{120} - 1)^2$. Consequently

$$1 - S(60, 49/2) \leq .07584, \quad 1 - T(120, 49/2) \leq .059062. \quad (6)$$

Both inequalities have strict arithmetic margins.

2.3 All integer lengths and the certificate

The initial ratio satisfies

$$\frac{\lambda(2b_*)}{\lambda(b_*)^2} > \frac{.9999865}{1.0013505^2} > c_* := .997290.$$

For an integer interval $32 \leq A \leq L \leq B$, the signed traces and (4) therefore give

$$T(L, b_*) \geq \frac{c_*^B(1 - .947852^A)}{(1 + .04236466 .872001^{A-32})^2}. \quad (7)$$

At $A = B = 32$, this proves the separate claim $\gamma_{32} > .06$.

For the longer scales, insert (6) into the coupled update (12) and the signed interpolation (14) below. As in version 1.3, every recurrence output is rounded upward to 12 significant decimal digits. At the seventh doubling the certified bounds are

$$n = 7680, \quad b = 3136, \quad p = 4.34388004302 \times 10^{-71}, \quad q = 1.23795864491 \times 10^{-97}.$$

Table 1 covers every integer from 33 through 11519. For each row its purity lower bound $\mathcal{K}$ satisfies $\mathcal{K} > 1/2$ and $E_m(7b/100) < \mathcal{K}/(1 - \mathcal{K})$, using the rational exponential upper bound (30). Thus the ground state is unique and $\gamma_L > .07$ on every listed interval.

Table 1: Version 1.4 finite certificate; both parities are included.

First length	Last length	Inverse temperature b
33	34	49/4
35	48	49/4
49	99	49/4
100	153	49/2
154	208	49/2
209	387	49
388	885	98
886	2295	196
2296	5534	392
5535	11519	784

For the infinite tail take $w_0 = 2 \times 10^{-97}$. The checker verifies $p^3 \leq w_0^2$, $q \leq w_0$, and the domain of the invariant (17). Its proof below applies unchanged: $w_{k+1} = 6w_k^2$ and $T(L, 3136 \cdot 2^k) \geq 1 - (15/2)w_k$ on $11520 \cdot 2^k \leq L \leq 23040 \cdot 2^k$. With $E = E_m(3136 \times 7/100)$, the two certified inequalities

$$E 6w_0 < 1, \quad E \frac{(15/2)w_0}{1 - (15/2)w_0} < 1$$

prove $\gamma_L > .07$ on this entire infinite tail by the argument following (18). The intervals meet without a missing integer, proving Theorem 1.

The checker `verify_spectral_refinement.py` performs 111 exact checks, including the filter, seed estimates, ten finite intervals, the length-32 consequence, and the infinite tail. It imports the directed rational arithmetic helpers of the retained version 1.3 checker. Numerical optimization selected the candidate filter only; the certificate contains fixed integer coefficients and requires no optimizer or floating-point acceptance decision.

3 Retained version 1.3 length-adaptive bound

Theorem 2. *Assume the signed spatial-transfer representation of [1, Proposition 3.1] and the periodic spectral estimates of [2, Proposition 3.4 and its proof], including their finite-input premises. Then*

$$\gamma_L > \frac{3}{50} = 0.06 \quad \text{for every integer } L \geq 34,$$

and the ground state is unique throughout this range. In particular, every odd $L \geq 35$ is included, and $\liminf_{L \rightarrow \infty} \gamma_L \geq 3/50$ over all integer lengths. For a positive exchange coupling J , multiply the bound by J .

3.1 Inputs and interpolation

Write $\lambda(b) > 0$ for the unique eigenvalue of largest modulus of the spatial operator, and define its residual absolute moment by

$$R_b(m) = \sum_{i \neq 0} \left| \frac{\mu_i(b)}{\lambda(b)} \right|^m.$$

All sums used here converge absolutely. Let $b_* = 49/4$. The proof of the cited companion proposition supplies

$$R_{b_*}(32) < .0442, \quad \rho_{b_*} < .874, \quad R_{2b_*}(32) < .9525^{32}, \quad (8)$$

$$\lambda(b_*) \leq 1.00139, \quad \lambda(2b_*)^{72} > .968, \quad (9)$$

$$S(60, 49/2) > .9002, \quad T(120, 49/2) > .9372. \quad (10)$$

Here ρ_b is the largest nondominant absolute eigenvalue ratio. These are periodic estimates, without boundary fields. The reseeding comparisons were checked by the version 1.1 certificate retained below. In particular, the rational number $c_* = .9967754$ satisfies

$$(c_* 1.00139^2)^{72} < .968, \quad \frac{\lambda(2b_*)}{\lambda(b_*)^2} > c_*.$$

Increasing moment orders in (8) gives, for every integer $L \geq 32$,

$$T(L, b_*) \geq \frac{c_*^L (1 - .9525^L)}{(1 + .0442 .874^{L-32})^2}. \quad (11)$$

The first residual bound uses the ratio ρ_{b_*} term by term. For the second, the sequence norm inequality gives $R_{2b_*}(L) \leq R_{2b_*}(32)^{L/32} < .9525^L$.

For the subsequent scales, suppose n is even and $1 - S(n, b) \leq p$, $1 - T(2n, b) \leq q$. With $f(z) = z^2/[2(1 - z)^2]$, put

$$a = 1 - (1 - p)^2(1 - q), \quad p_+ \geq f(a), \quad q_+ \geq f(1 - (1 - q)^2(1 - p_+)). \quad (12)$$

The coupled update, proved in Lemma 2 below, bounds the defects at the next scale $(2n, 2b)$ by p_+, q_+ . Cancellation also gives $S(n, 2b) \geq 1 - a$ at the *same* reference length. When $p, a < 1/2$, the concentration lemma below yields

$$R_b(n) \leq d(p), \quad R_{2b}(n) \leq d(a), \quad d(z) = \frac{z}{2 - 3z}.$$

The leading spatial weight at $(2n, 2b)$ is at least $1 - p_+$. Since $Z_{2n}(b) \geq \lambda(b)^{2n}$, it follows that

$$\left(\frac{\lambda(2b)}{\lambda(b)^2} \right)^{2n} \geq (1 - p_+) \frac{Z_{2n}(2b)}{Z_{2n}(b)^2} \geq (1 - p_+)(1 - q) =: g. \quad (13)$$

Choose rational numbers $0 < c, \rho_0, \rho_1 < 1$ with $c^{2n} \leq g$, $\rho_0^n \geq d(p)$, and $\rho_1^n \geq d(a)$. Each residual eigenvalue ratio is bounded by the corresponding ρ . For any integer interval $n \leq A \leq L \leq B$, the signed trace therefore gives

$$T(L, b) \geq \frac{c^B (1 - d(a) \rho_1^{A-n})}{(1 + d(p) \rho_0^{A-n})^2} =: \mathcal{K}(A, B; b), \quad (14)$$

provided the displayed numerator factor is positive. No parity restriction is imposed on L . The numerator uses the lower bound on the signed residual, and the denominator uses its upper bound.

3.2 Finite coverage

Starting with $n = 60$, $b = 49/2$, $p = .0998$, and $q = .0628$, the checker iterates (12), rounding each output upward to 12 significant decimal digits. The rounded p_+ is used in the second update. At the seventh doubling it obtains

$$n = 7680, \quad b = 3136, \quad p = 2.30218227586 \times 10^{-56}, \quad q = 1.75070874875 \times 10^{-83}. \quad (15)$$

All these decimals are exact upper bounds. The full sequence and rational interpolation parameters are in the machine-readable certificate.

For each interval in Table 2, the checker uses (11) or (14) to certify

$$\mathcal{K} > \frac{1}{2}, \quad E_m(3b/50) < \frac{\mathcal{K}}{1 - \mathcal{K}}, \quad (16)$$

where E_m is the rational exponential upper bound in (30). For the initial four intervals, the lower bound uses c_*^B , $.9525^A$, and $.0442.874^{A-32}$. The largest physical Gibbs weight exceeds $1/2$, so the ground state is unique; the ratio of a next-energy weight to it is at most $(1 - \mathcal{K})/\mathcal{K}$. Thus (16) proves $\gamma_L > 3/50$ for every integer $34 \leq L \leq 11519$.

Table 2: Certified finite intervals; every integer in each row is covered.

First length	Last length	Inverse temperature b
34	35	49/4
36	46	49/4
47	84	49/4
85	116	49/4
117	164	49/2
165	202	49/2
203	302	49
303	421	49
422	814	98
815	1937	196
1938	4748	392
4749	11519	784

The root calculations in parameter selection are proposals only. Every root direction is certified by an integer-power comparison over the rationals. Positive products in those comparisons and in (14) are rounded outward to multiples of 10^{-140} . Floor or ceiling of the integer quotient gives the corresponding enclosure; induction through exponentiation by squaring preserves its direction. The chosen interval covering is then checked for both its inequalities and the absence of missing integers.

3.3 An asymmetric invariant for the infinite tail

Suppose

$$p^{3/2} \leq w, \quad q \leq w, \quad 0 < w \leq 10^{-6}. \quad (17)$$

Write $x = w^{1/3} \leq 1/100$, so $p \leq x^2$ and $q \leq x^3$. Since $a \leq 2p + q \leq (2 + x)x^2$, the exact scalar comparisons

$$\frac{(2.01)^2}{2(1 - 2.01 \times 10^{-4})^2} < 3, \quad \frac{(2.03)^2}{2(1 - 2.03 \times 10^{-6})^2} < 3$$

imply, using the unrounded upper updates,

$$p_+ \leq 3x^4, \quad q_+ \leq f(2x^3 + 3x^4) \leq 3w^2.$$

Consequently $p_+^{3/2} \leq \sqrt{27} w^2 < 6w^2$ and $q_+ \leq 6w^2$. The invariant propagates with $w_+ = 6w^2$. For every integer $3n/2 \leq L \leq 3n$, increasing the absolute-moment order gives

$$R_b(L) \leq d(p)^{3/2} \leq w, \quad R_{2b}(L) \leq d(a)^{3/2} \leq (2.01)^{3/2} w < 3w.$$

We used $d(z) \leq z$ for $0 \leq z \leq 1/3$; both p and a lie in this domain. Also $g \geq 1 - q - p_+ \geq 1 - 1.03w$. By (13), the leading-eigenvalue contribution at length L is at least $g^{L/(2n)} \geq g^2 \geq 1 - 2.06w$. It follows that

$$T(L, b) \geq \frac{(1 - 2.06w)(1 - 3w)}{(1 + w)^2} \geq 1 - 7.06w \geq 1 - \frac{15}{2}w > \frac{1}{2}. \quad (18)$$

Here $(1 + w)^{-2} \geq 1 - 2w$, and multiplication of the three nonnegative factors bounds their defect by the sum of the individual defects.

At (15), take $w_0 = 2 \times 10^{-83}$. The checker verifies $p^3 \leq w_0^2$, $q \leq w_0$, and $6w_0 < 1$. Let

$$n_k = 7680 \cdot 2^k, \quad b_k = 3136 \cdot 2^k, \quad w_k = \frac{1}{6}(6w_0)^{2^k}, \quad B_0 = 15/2.$$

The invariant and (18) apply on every interval $[11520 \cdot 2^k, 23040 \cdot 2^k]$. On each interval the ground state is unique, and

$$e^{-b_k \gamma_L} \leq \frac{B_0 w_k}{1 - B_0 w_k} \leq \frac{B_0}{6(1 - B_0 w_0)} (6w_0)^{2^k}.$$

Set $E = E_m(3136 \times 3/50) \geq e^{3136(3/50)}$. The final two exact exponential checks are

$$E \cdot 6w_0 < 1, \quad E \frac{B_0 w_0}{1 - B_0 w_0} < 1.$$

Therefore

$$e^{b_k(3/50)} e^{-b_k \gamma_L} \leq \frac{B_0}{6(1 - B_0 w_0)} (E \cdot 6w_0)^{2^k} < 1.$$

This proves the strict target bound for every integer $L \geq 11520$. Together with the finite coverage, it proves Theorem 2.

The certificate `verify_adaptive_bound.py` implements 88 exact checks. Its analytic assumptions and the proof above remain outside machine-formal verification. The gain comes from retaining unequal defects, changing the interpolation intervals, and using an invariant matched to their different decay rates. No extra finite input is used.

4 Including odd lengths

This section retains the version 1.2 argument and its smaller constant.

Theorem 3. *Assume the signed spatial-moment representation of [1, Proposition 3.1] and the periodic concentration estimates, positive dominant eigenvalues, and adjacent-temperature eigenvalue ratios of [2, Proposition 3.4], including their finite-input premises. The periodic chain has a unique ground state and*

$$\gamma_L > \frac{\log(125/39)}{49} > 0.02377 \quad \text{for every integer } L \geq 120.$$

In particular, this holds for every odd $L \geq 121$, and $\liminf_{L \rightarrow \infty} \gamma_L \geq \log(125/39)/49$ over all integer lengths. For a positive exchange coupling J , multiply the bounds by J .

We first record the concentration bound on the domain needed below. The companion states its rational version for defects at most .1; its elementary proof extends to all defects below 1/2.

Lemma 1 (Concentration). *Let (p_i) be a finite or countable probability list with $\sum_i p_i^2 \geq 1 - z$, where $0 \leq z < 1/2$. Its largest weight p_0 is unique, and*

$$\frac{\sum_{i \neq 0} p_i}{p_0} \leq d(z), \quad d(z) = \frac{z}{2 - 3z}.$$

Proof. A summable nonzero nonnegative list attains its maximum. Since $\sum_i p_i^2 \leq p_0$, one has $p_0 \geq 1 - z > 1/2$. Put $x = 1 - p_0 \leq z$. Then $2x(1 - x) \leq 1 - \sum_i p_i^2 \leq z$, so $x \leq z/[2(1 - x)] \leq z/[2(1 - z)]$. Applying the increasing function $x \mapsto x/(1 - x)$ gives the result; the displayed upper bound for x is below one when $z < 1/2$. $\square$

Proof of Theorem 3. Reindex the companion's scales $j \geq 2$ by $k = j - 2 \geq 0$, and set

$$n_k = 120 \cdot 2^k, \quad \beta_k = 49 \cdot 2^k, \quad u_k = \frac{1}{6} \left(\frac{39}{125} \right)^{2^k}. \quad (19)$$

Thus $u_0 = 13/250$ and $u_{k+1} = 6u_k^2 \leq u_k$. Write $n = n_k$, $\beta = \beta_k$, $u = u_k$, and $v = u_{k+1}$. The cited propositions supply real spatial eigenvalues $\mu_i(b)$ with absolutely convergent moments and

$$Z_L(b) = \sum_i \mu_i(b)^L \quad (\text{every integer } L \geq 4).$$

At $b = \beta, 2\beta$, their respective unique eigenvalues of maximal modulus are positive; denote these by λ, λ' . With $c = \lambda'/\lambda^2$, the companion gives

$$S(n, \beta) \geq 1 - u, \quad T(2n, \beta) \geq 1 - u, \quad 0 < c \leq 1, \quad c^{2^n} \geq (1 - u)(1 - v). \quad (20)$$

Positivity of the leading eigenvalue does not assume an odd-chain gap: the companion proves it from positivity of physical partition functions and dominance in the signed trace at arbitrarily large odd lengths.

The necessary control at the *same* reference length n and the next temperature follows by partition-function cancellation:

$$S(n, 2\beta) = \frac{S(n, \beta)^2 T(2n, \beta)}{T(n, \beta)^2} \geq (1 - u)^3 \geq 1 - 3u. \quad (21)$$

Here $T(n, \beta) \leq 1$. Because n is even, the two spatial lists at this length are probability lists. Lemma 1, applicable since $3u \leq 39/250 < 1/2$, gives

$$\sum_{i \neq 0} \left| \frac{\mu_i(\beta)}{\lambda} \right|^n \leq d(u), \quad \sum_{i \neq 0} \left| \frac{\mu_i(2\beta)}{\lambda'} \right|^n \leq d(3u). \quad (22)$$

Each absolute eigenvalue ratio is at most one. Increasing its exponent from n to any integer $L \geq n$ therefore decreases every summand. Absolute convergence and the triangle inequality yield

$$Z_L(\beta) = \lambda^L (1 + \epsilon_L), \quad |\epsilon_L| \leq d(u), \quad Z_L(2\beta) = (\lambda')^L (1 + \epsilon'_L), \quad |\epsilon'_L| \leq d(3u).$$

These estimates allow negative nondominant eigenvalues. No probability interpretation of an odd spatial moment is used. In particular, (21) is essential: a higher-moment upper bound alone cannot bound a lower moment by taking a root.

For every integer $L \in [n, 2n]$, $c^L \geq c^{2^n}$, and hence

$$T(L, \beta) \geq K(u) := \frac{(1 - u)(1 - 6u^2)(1 - d(3u))}{(1 + d(u))^2} \geq 1 - \frac{9}{2}u. \quad (23)$$

To prove the last inequality on the entire interval $0 \leq u \leq 13/250$, use the exact identity

$$K(u) - \left(1 - \frac{9}{2}u\right) = \frac{u[4 - 60u + 243u^2(1 - 2u) + 324u^4]}{2(2 - 9u)(1 - u)} \geq 0. \quad (24)$$

Indeed, $4 - 60u \geq 22/25 > 0$, $1 - 2u > 0$, and both denominator factors are positive on that interval. In particular, $T(L, \beta) \geq 383/500 > 1/2$.

Let w_0 be the largest physical Gibbs weight, counting eigenvectors with multiplicity. Since $w_0 \geq T(L, \beta) > 1/2$, the ground state is unique. For an eigenvector of the next distinct energy,

$$\begin{aligned} e^{-\beta_k \gamma_L} &= \frac{w_1}{w_0} \leq \frac{(9/2)u_k}{1 - (9/2)u_k} \\ &\leq \frac{375}{383} \left(\frac{39}{125}\right)^{2^k} < \left(\frac{39}{125}\right)^{2^k}. \end{aligned}$$

Taking logarithms and dividing by $\beta_k = 49 \cdot 2^k$ proves the claimed strict bound on each interval $[120 \cdot 2^k, 240 \cdot 2^k]$. These intervals cover all integer $L \geq 120$. The rounded consequence follows from the exact exponential comparison described in (30). The liminf inequality follows from the uniform finite-length bound. $\square$

4.1 Certificate and contribution

The script `verify_signed_extension.py` makes 21 exact checks: the polynomial coefficient identity in (24), endpoint inequalities proving its sign over the full interval, the purity floor, the strict prefactor $375/383 < 1$, and the comparison with .02377. It uses no sampling grid or numerical logarithm in a decision. The concentration and signed-trace steps above are analytic proofs, not machine-formalized theorems.

This version adds the signed-moment interpolation argument to the even-length corollary retained below. It uses more of the companion's already proved spectral information and requires no new thermal or variational input. The argument is conditional on those results; it is not an independent proof of the original Haldane-gap claim. No exhaustive priority search for this parity extension has been made.

5 The retained version 1.1 corollary

Despite its boundary-field title, Section 3 of [2] uses the periodic partition function $Z_n(b)$ of precisely the Hamiltonian above, without endpoint fields. Its separate boundary partition functions do not enter the following argument.

Theorem 4. *Assume the spatial-moment representation of [1, Proposition 3.1] and the periodic concentration estimates of [2, Proposition 3.4], including their finite-input premises. The periodic chain has a unique ground state and*

$$\gamma_L > \frac{\log(125/39)}{49} > 0.02377 \quad \text{for every even } L \geq 120.$$

Consequently $\liminf_{L \rightarrow \infty, L \text{ even}} \gamma_L \geq \log(125/39)/49$. For a positive exchange coupling J , multiply these bounds by J .

Proof. The companion sets $b_j = 2^j(49/4)$ and, for $j \geq 2$, $n_j = 120 \cdot 2^{j-2}$, $M_j = 2n_j$, and $u_2 = .052$. Its Proposition 3.4 gives $1 - S(n_j, b_j) \leq u_j$ and $1 - T(M_j, b_j) \leq u_j$. In particular, $j = 2$ gives

$$1 - S(120, 49) \leq \frac{13}{250}, \quad 1 - T(240, 49) \leq \frac{13}{250}. \quad (25)$$

Apply [1, Proposition 4.3] with the new starting parameters

$$n_0 = 120, \quad \beta_0 = 49, \quad u_0 = \frac{13}{250}, \quad C = 6, \quad r = Cu_0 = \frac{39}{125}.$$

Here β_0 is the companion's b_2 , not its b_0 . Writing $G(u) = \frac{1}{2}(\sum_{k=1}^3(1-u)^{-k})^2$, exact substitution gives

$$0 < u_0 < \frac{1}{6}, \quad G(u_0) = \frac{989224080031250}{177210755074809} < 6, \quad r < 1, \quad C(1 - 3u_0) = \frac{633}{125} > 3.$$

The two seed assumptions and every scalar hypothesis of that proposition therefore hold. Its interpolation step covers all even $L \geq n_0$, and its conclusion is uniqueness and $\gamma_L > \log(1/r)/\beta_0 = \log(125/39)/49$. The strict comparison with .02377 follows from the rational Taylor upper bound in (30), which verifies $e^{49(2377/100000)} < 125/39$. $\square$

5.1 Scalar certificate and attribution

The script `verify_companion_corollary.py` checks the substitution above and the reseeding comparisons in the proof of [2, Proposition 3.4]. That proof starts from the finite periodic inputs of [1, Corollary 5.6], including the 72- and 120-site trial bounds already covered by our recorded finite verification. It requires no new boundary-field thermal calculation for the periodic seed used here.

In the companion's notation, the checker verifies the rational majorants $D(32) < .0442$ and $D(120) < .00000025$, the residual spectral ratio bound, and the seeds $S(60, 49/2) > .9002$ and $T(120, 49/2) > .9372$. The coupled update then gives defects below .0502 and .0198, both below .052 at the scales in (25). Cube-root bounds are checked by cubing positive rational majorants; no floating-point root or logarithm determines a pass/fail decision.

The underlying reseeding argument and uniform-gap criterion are both OpenAI's published results. Our contribution here is their explicit combination, its scalar certificate, and the resulting comparison. The value $\log(125/39)/49 = 0.023770450840258259\dots$ was extracted during our prior-art review; we have not located it explicitly stated in the examined sources. This limited observation does not establish priority. Version 1.0's comparison omitted this consequence of the companion paper; the present version corrects that omission.

6 The retained version 1.0 refinement

The earlier, weaker result remains a separate reproducible extension of the periodic paper's final seed pair. Its dependencies on [1] are the following two inputs:

1. For each $b > 0$, a real finite or countable sequence $\lambda_i(b)$ obeys $\sum_i |\lambda_i(b)|^4 < \infty$ and $Z_L(b) = \sum_i |\lambda_i(b)|^L$ for every even $L \geq 4$ (Proposition 3.1).
2. The initial estimates are

$$1 - S(2304, 784) \leq \frac{84513}{10^7}, \quad 1 - T(4608, 784) \leq \frac{27493}{5 \cdot 10^6} \quad (26)$$

(Proposition 5.5, including its finite certificates).

All references are to the fixed upstream revision listed at the end.

Theorem 5. *Under these inputs, the ground state is unique and*

$$\gamma_L > \frac{47}{10000} \quad \text{for every even } L \geq 2304.$$

Consequently $\liminf_{L \rightarrow \infty, L \text{ even}} \gamma_L \geq 47/10000$. For any positive exchange coupling J , multiply these bounds by J .

The argument below rederives the needed bootstrap steps of [1, Lemma 4.2 and Proposition 4.3]. It does not establish the two inputs independently of the upstream analytic argument.

7 Purity estimates and gap extraction

For an even reference length $n \geq 4$, both quantities in (1) are purities of probability distributions, so they belong to $(0, 1]$. For spatial purity use $|\lambda_i(b)|^n/Z_n(b)$; for physical purity use the Gibbs weights, counting each eigenvector with multiplicity. If $P = \sum_i w_i^2$, the normalized squares have purity P' satisfying

$$1 - P' = \frac{2 \sum_{i < k} w_i^2 w_k^2}{P^2} \leq \frac{2(\sum_{i < k} w_i w_k)^2}{P^2} = f(1 - P), \quad f(x) = \frac{x^2}{2(1 - x)^2}.$$

Here f is increasing on $[0, 1]$.

Lemma 2 (Rounded update). *For even $n \geq 4$ and $b > 0$, if $1 - S(n, b) \leq p$ and $1 - T(2n, b) \leq q$, with $p, q \in [0, 1]$, any $p_+, q_+ \in [0, 1]$ satisfying*

$$p_+ \geq f(1 - (1 - p)^2(1 - q)), \quad q_+ \geq f(1 - (1 - q)^2(1 - p_+)) \quad (27)$$

bound $1 - S(2n, 2b)$ and $1 - T(4n, 2b)$ respectively.

Proof. Cancellation of partition functions gives

$$S(n, 2b) = \frac{S(n, b)^2 T(2n, b)}{T(n, b)^2}, \quad T(4n, b) = \frac{T(2n, b)^2 S(2n, 2b)}{S(2n, b)^2}.$$

The denominators are at most one. Apply the probability-squaring inequality first in the spatial variable and then in the temperature variable. Use the already rounded p_+ in the second application. $\square$

Lemma 3 (Intermediate lengths). *Under the same hypotheses, put $K = (1 - p)^2(1 - q)$. If $K > 1/2$, every even $L \in [n, 2n]$ has a unique ground state and*

$$\gamma_L \geq \frac{1}{b} \log \frac{K}{1 - K}. \quad (28)$$

Proof. The monotonicity of sequence ℓ^r norms, applied to the spatial moments at b and $2b$, yields

$$Z_L(b) \leq Z_n(b)^{L/n}, \quad Z_L(2b) \geq Z_{2n}(2b)^{L/(2n)}.$$

Hence

$$T(L, b) \geq [T(2n, b) S(n, b)^2]^{L/(2n)} \geq K.$$

The last step uses a base in $(0, 1]$ and an exponent at most one. The largest Gibbs weight w_0 is at least the purity, because $\sum_i w_i^2 \leq (\max_i w_i) \sum_i w_i$. Thus $w_0 > 1/2$, excluding ground-state degeneracy. For one eigenvector at the next energy,

$$e^{-b\gamma_L} = \frac{w_1}{w_0} \leq \frac{1 - w_0}{w_0} \leq \frac{1 - K}{K}.$$

Taking logarithms proves the assertion. $\square$

Table 3: Upward-rounded purity-defect bounds.

j	n_j	b_j	p_j	q_j
0	2304	784	0.0084513	0.0054986
1	4608	1568	2.59×10^{-4}	6.45×10^{-5}
2	9216	3136	1.70×10^{-7}	8.35×10^{-9}
3	18432	6272	6.07×10^{-14}	1.40×10^{-16}
4	36864	12544	7.39×10^{-27}	3.93×10^{-32}
5	73728	25088	1.10×10^{-52}	3.09×10^{-63}

8 A finite certificate and the infinite tail

Use the exact rational numbers in Table 3 with $n_j = 2304 \cdot 2^j$ and $b_j = 784 \cdot 2^j$. Every displayed decimal denotes an exact rational, not a floating-point approximation. The script `verify_gap_improvement.py` checks (27) at each of the five steps.

For $j = 0, \dots, 4$, set $K_j = (1 - p_j)^2(1 - q_j)$. The certificate verifies

$$K_j > \frac{1}{2}, \quad e^{(47/10000)b_j} < \frac{K_j}{1 - K_j}. \quad (29)$$

Lemma 3 therefore proves the strict target bound throughout $2304 \leq L \leq 73728$, restricted to even lengths.

For clarity, the exponential comparisons use no numerical transcendental evaluation. Given rational $x \geq 0$, choose $m = 4\lfloor x \rfloor + 64$. Since $\rho = x/(m+2) < 1$, the Taylor series and the ratio bound for its remaining terms imply

$$e^x \leq E_m(x) := \sum_{k=0}^m \frac{x^k}{k!} + \frac{x^{m+1}}{(m+1)!} \frac{1}{1 - x/(m+2)}. \quad (30)$$

Every term on the right is rational. The script verifies the stronger inequality obtained by replacing e^x with $E_m(x)$ in (29).

The final row controls all remaining lengths. Let

$$u = 1.10 \times 10^{-52}, \quad C = 5, \quad r = Cu, \quad G(v) = \frac{1}{2}((1-v)^{-1} + (1-v)^{-2} + (1-v)^{-3})^2.$$

Exact checks give $0 < u < 1/6$, $G(u) < C$, $r < 1$, and $C(1 - 3u) > 3$. Define $u_k = C^{-1}r^{2^k}$ for $k \geq 0$. Both defects at $(2^k n_5, 2^k b_5)$ are bounded by u_k : indeed, $u_{k+1} = Cu_k^2 < u_k$, and

$$f(1 - (1 - u_k)^3) = G(u_k)u_k^2 \leq Cu_k^2.$$

Lemma 2 then applies to both coordinates by induction. For even $L \in [2^k n_5, 2^{k+1} n_5]$, the proof of Lemma 3 gives $w_0 \geq (1 - u_k)^3 \geq 1 - 3u_k$. Consequently

$$e^{-(2^k b_5) \gamma_L} \leq \frac{3u_k}{1 - 3u_k} \leq \frac{3}{C(1 - 3u)} r^{2^k} < r^{2^k}.$$

These intervals cover every even $L \geq n_5$, proving

$$\gamma_L > \frac{\log(1/(Cu))}{b_5} = \frac{\log(10^{53}/55)}{25088} = 0.004704626783460297 \dots > \frac{47}{10000}. \quad (31)$$

The final strict comparison is certified by $E_m((47/10000)25088) < 10^{53}/55$, again over the rationals. Together with the five finite intervals this proves Theorem 5. A strict bound for each finite length yields the stated non-strict bound on the liminf.

9 Reproducibility and scope

The version 1.0 quantitative gain comes from retaining the unequal seed bounds and performing a few more coupled updates before using a common envelope. The upstream theorem instead rounds both seed defects to $1/100$ and uses $C = 5$, giving $\log(20)/784$. No new thermal or variational input is required for this refinement. This is a comparison with that paper’s stated constant, not with every consequence of the published pair of papers. Theorem 4 is stronger in both constant and length range. We make no priority or optimality claim.

Theorem 3 further removes the parity restriction without changing the version 1.1 constant. Theorem 2 improves both the constant and the starting length using the same finite inputs. Theorem 1 then sharpens the spectral inputs and raises the bound again, without changing the finite computation data. The accompanying repository contains all five exact certificates, the upstream source and verification material at the fixed commit, fresh verification results, and `independent_audit.py`. The latter reconstructs selected computations from the printed definitions and tables without importing upstream verifier functions. The recorded audit, completed on 7 October 2026, comprises:

- A full fresh run of the upstream verifier, with all checks passing and all finite results newly recomputed. It reproduces 27 thermal certificates, the 72- and 120-site variational contractions, and the associated arithmetic and scalar checks.
- A separate arbitrary-precision integer reconstruction of both variational contractions. The complete numerator and norm integers match the upstream evidence. A direct four-site wavefunction calculation also agrees with the contraction formula.
- A separate full-spectrum diagonalization on chains of four to eight sites, including twists, in 16 cases. All values lie within the printed tolerances. This floating-point calculation is a diagnostic; it does not provide rigorous thermal enclosures.

The full upstream run took approximately 22 minutes with six requested workers, Python 3.12.14, NumPy 2.3.5, and SciPy 1.17.0. All five bound certificates need only Python’s standard library. Run `make check` with assertions enabled to check all five. For the new spectral-refinement certificate alone, run

```
python3 verify_spectral_refinement.py
```

Rational comparisons determine all numerical pass/fail decisions; decimal logarithms and floating-point margins are display-only. The companion’s selected analytic sources are preserved unmodified under `source_boundary/`, with Git blob identities recorded in `boundary_source_manifest.json`.

Neither successful execution nor this note is a formal verification of the upstream analytic assumptions. No claim is made here for the same numerical bound at lengths below 33, arbitrary open boundaries, or uniqueness of all infinite-volume ground states. The constants are conditional lower bounds, not estimates of the actual physical gap.

Autonomous authorship and human contribution. GPT-6 Astra Max autonomously selected this result for investigation, examined the upstream argument, derived the refined bound, implemented and ran the checks, and wrote this manuscript. During the user-requested prior-art review, the agent identified the stronger corollary, verified its parameter substitution, and prepared version 1.1. When the user asked about odd lengths, the agent derived the signed-moment extension, implemented its certificate, and wrote version 1.2. During the requested continued investigation, the agent developed the length-dependent temperature choice and asymmetric tail invariant, checked the new certificate, and wrote version 1.3. Continuing the investigation, the agent selected a tighter filter, derived the refined spectral estimates and

seeds, implemented the 0.07 certificate, and wrote version 1.4. The human user initiated the open-ended investigation, authorized computation and publication, and requested attribution and layout revisions; the mathematical extension and its verification code were produced by the agent. The human also requested the prior-art review and asked that newly verified findings be incorporated into the repository. The model designation GPT-6 Astra Max was supplied by the human user; the execution agent was Codex. A backend model snapshot identifier was not available in the session. The upstream mathematical construction, reseeding argument, and input certificates are credited to [1, 2]. This is an unreviewed technical preprint; no independent human expert review has been recorded. Publication through the owner’s account does not make this an official OpenAI publication or endorsement. Corrections are welcome through the repository’s issue tracker.

References

- [1] OpenAI, *The periodic spin-one Haldane gap*, 24 September 2026, OpenAI Math Release preprint. Repository <https://github.com/openai/math>, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. File `preprints/The-periodic-spin-one-Haldane-gap-September-24-2026/paper.pdf`. The companion analytic source and finite verifier are at the same revision.
- [2] OpenAI, *A boundary-field gap for the spin-one Heisenberg chain*, 24 September 2026, OpenAI Math Release preprint. Repository <https://github.com/openai/math>, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. File `preprints/A-boundary-field-gap-for-the-spin-one-Heisenberg-chain-September-24-2026/paper.pdf`. Section 2 fixes the periodic normalization; Proposition 3.4 supplies the periodic concentration estimates used here.